$F = \sqrt{T^2 + W^2} = \sqrt{8.0^2 + (0.30 \times 9.81)} = 8.52\,\text{N}$

11 A At point P,

Gravitational field strength due to M = $\frac{GM}{x^2}$

Gravitational field strength due to $9M = \frac{G9M}{y^2}$

As resultant gravitational field strength = 0,

$$\frac{GM}{x^2} = \frac{G9M}{y^2}$$

$$\frac{1}{x^2} = \frac{9}{y^2} \Rightarrow x : y = 1 : 3$$

12 D